

4

MEASUREMENT AND SPACE

GEOMETRY

Geometry is all around us. Designs and patterns involving angles, parallel lines, triangles and quadrilaterals can be found everywhere, in our homes, on transport, in construction, art and nature. This scene from the National Museum of Australia in Canberra shows the importance of angles, lines and shapes in architecture and design.



Chapter outline

	Working mathematically				
4.01 Angle geometry	U	F		R	C
4.02 Angles on parallel lines	U	F		R	C
4.03 Classifying triangles	U	F		R	C
4.04 Classifying quadrilaterals	U	F		R	C
4.05 Properties of quadrilaterals	U	F		R	C
4.06 Angle sums of triangles and quadrilaterals	U	F		R	C
4.07 Angle sum of a polygon*	U	F		R	C
4.08 Networks*	U	F			C

*YEAR 9 EXTENSION

Wordbank

angle sum The total of the sizes of the angles in a shape, such as a triangle

bisect To cut in half

convex quadrilateral A quadrilateral whose vertices all point outwards.

diagonal An interval joining 2 non-adjacent vertices of a shape

exterior angle of a triangle An 'outside' angle of a triangle formed after extending one of the sides of the triangle

scalene triangle A triangle with no equal sides

supplementary angles 2 angles that add to 180°

Quiz
Wordbank 4

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (12):

4.01 Angles at a point and vertically opposite angles • Angle geometry

4.02 Angles on parallel lines • Angle relationships • Angle geometry

4.03 Properties of triangles • Classifying triangles

4.04, 4.05 Properties of quadrilaterals • Classifying quadrilaterals

4.06 Angle sums of triangles and quadrilaterals • Angles in polygons • Exterior angle of a triangle

4.08 Networks

Twig videos (2)

4.08 The seven bridges of Konigsberg • Networks: Labyrinths and mazes

Quizzes (6):

- Wordbank 4
- SkillCheck 4
- Mental skills 4A
- Mental skills 4B
- Language of maths 4
- Test yourself 4

Skillsheets (4):

SkillCheck Types of angles • Line and rotational symmetry

4.02 Angles and parallel lines

4.03 Naming shapes

Worksheets (23):

SkillCheck A page of angles • Symmetry

4.01 Straight angles, right angles and revolutions

4.02 Find the unknown angle 1 • Investigating angles on parallel lines • Angles in parallel lines • What is the diagram? • Matching angles

4.03 Properties of triangles • Triangle geometry

4.04 Classifying quadrilaterals • Properties of quadrilaterals • Always, sometimes, never true • Classifying triangles and quadrilaterals

4.05 Properties of quadrilaterals • Diagonal properties of quadrilaterals • Always, sometimes, never true

4.06 Angles in triangles • Find the unknown angle 2 • Deductive geometry • Shapes and angles review

4.07 Angle sum of a polygon • Angles in regular polygons • Equal angles

Mind map: Geometry

Puzzles (12):

SkillCheck Angle cards

4.01 Angles: A dog day • Angles

4.02 Angles in parallel lines

4.03 Classifying triangles

4.04 What shape am I? • Singing in the car

4.06 Mixed angles • Angles in triangles • Triangles and quadrilaterals

4.07 Shape puzzle 1

Language of maths Geometry crossword

Presentation (1):

4.05 Angles and shapes



Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- solve geometrical problems involving angles on a straight line, angles at a point and vertically opposite angles
- solve geometrical problems involving corresponding, alternate and co-interior angles on parallel lines crossed by a transversal
- classify triangles according to their side and angle properties
- identify convex and non-convex quadrilaterals
- classify squares, rectangles, rhombuses, parallelograms, kites and trapeziums

- solve geometrical problems involving the angle sum of a triangle and quadrilateral and the exterior angle of a triangle
- (EXTENSION, YEAR 9 PATH) solve problems involving the angle sum of a polygon
- (EXTENSION, YEAR 9 PATH) identify the parts of a network and investigate Euler's rule

SkillCheck ANSWERS ON P. 579

1 Draw 2 different examples of each type of angle.

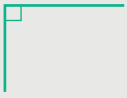
a acute angle **b** right angle **c** obtuse angle **d** reflex angle

2 Classify each angle size as being acute, obtuse, right, reflex, straight or a revolution.

a 25° **b** 100° **c** 300° **d** 128°

e 90° **f** 360° **g** 286° **h** 180°

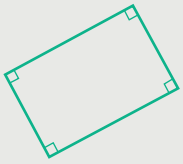
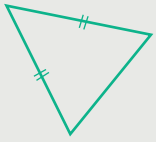
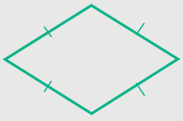
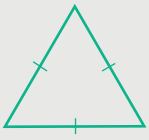
3 Name each type of angle(s) marked.

a  **b**  **c** 

4 Find the value of x in each equation.

a $x + 30 = 90$ **b** $x + 57 = 180$ **c** $x + 121 + 77 = 360$

5 Copy each shape, name it and draw all axes of **symmetry**.

a  **b**  **c**  **d** 

6 Write the order of **rotational symmetry** of each shape in question **5**.

7 Which quadrilateral has all 4 sides equal and all 4 angles equal?
Select the correct answer **A**, **B**, **C** or **D**.

A parallelogram **B** rhombus **C** rectangle **D** square

8 Draw a scalene triangle.

9 **a** Draw a parallelogram and its diagonals.

b Are the lengths of the diagonals of a parallelogram equal?


Quiz
SkillCheck 4


Skillsheets
Types of angles

Line and rotational symmetry

Worksheets
A page of angles

Symmetry

Puzzle
Angle cards

4

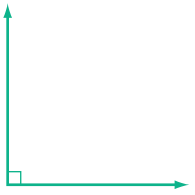
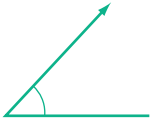
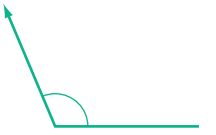
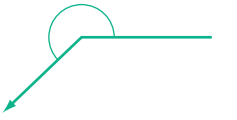
4.01 Angle geometry

Worksheet
Straight angles, right angles and revolutions

Puzzles
Angles: A dog day

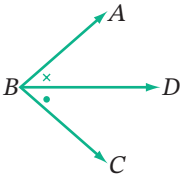
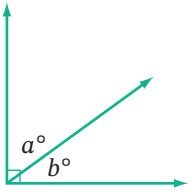
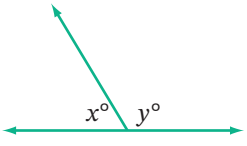
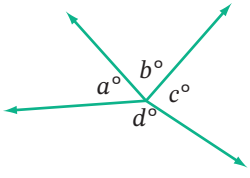
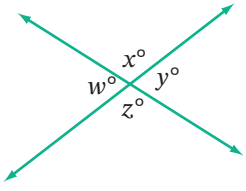
Angles

Classifying angles

Right angle	Straight angle	Revolution
90° (quarter-turn) 	180° (half-turn) 	360° (complete turn) 
Acute angle	**Obtuse angle**	**Reflex angle**
Less than 90° 	Between 90° and 180° 	Between 180° and 360° 

Video
Angles at a point and vertically opposite angles

Angle facts

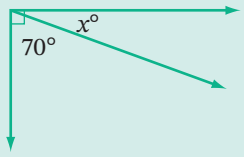
Adjacent angles	Complementary angles	Angles in a right angle
Angles next to each other, sharing a common arm. $\angle ABD$ and $\angle DBC$ are adjacent. 	2 angles that have a sum of 90°, for example, 35° and 55°.	Are complementary. $a + b = 90$ 
	Supplementary angles	
	2 angles that have a sum of 180°, for example, 140° and 40°.	
Angles on a straight line	**Angles at a point**	**Vertically opposite angles**
Are supplementary. $x + y = 180$ 	(In a revolution) Add up to 360°. $a + b + c + d = 360$ 	Are equal. $w = y$ and $x = z$ 

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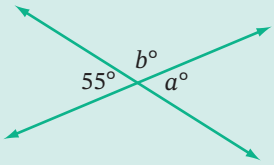
Example 1

Find the value of each variable, giving reasons.

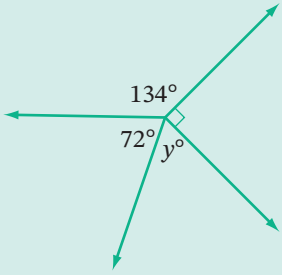
a



b



c



Solution

a $x + 70 = 90$ (Angles in a right angle)

$x = 90 - 70$

$= 20$

The reason is written inside brackets.

b $a = 55$ (Vertically opposite angles.)

$b + 55 = 180$ (Angles in a straight line)

$b = 180 - 55$

$= 125$

c $y + 72 + 134 + 90 = 360$ (Angles at a point)

$y + 296 = 360$

$y = 360 - 296$

$= 64$

Video
Angle geometry

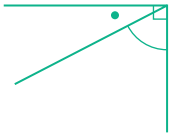
4.01

EXERCISE 4.01 ANSWERS ON P. 580

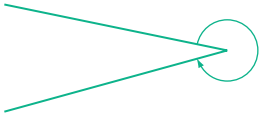
Angle geometry U F R C

1 Classify each type of angle(s). C

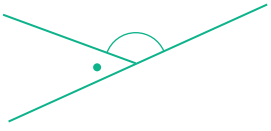
a



b



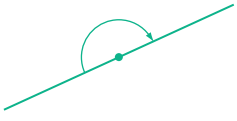
c



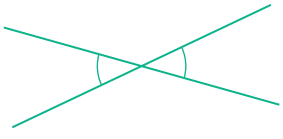
d



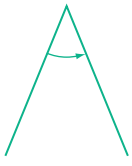
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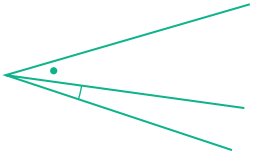
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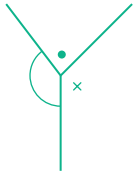
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h



i



Foundation | Standard | Complex

2 For each angle size, find the complementary angle.

a 16° **b** 33° **c** 71° **d** 2°

3 For each angle size, find the supplementary angle.

a 25° **b** 149° **c** 107° **d** 48°

4 What is the sum of the angles at a point? Select the correct answer **A**, **B**, **C** or **D**.

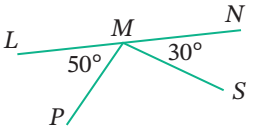
A 90° **B** 180° **C** 270° **D** 360°

EXAMPLE 1

5 Find size of $\angle SMP$. Select **A**, **B**, **C** or **D**.

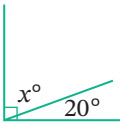
A 80° **B** 280°

C 100° **D** 10°

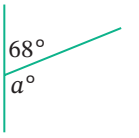


6 Find the value of each variable, giving reasons. R C

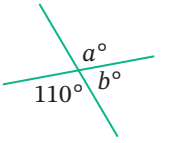
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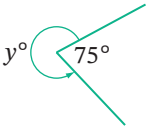
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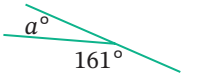
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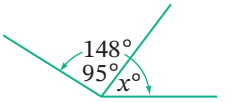
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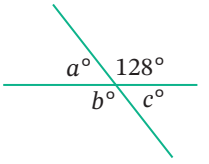
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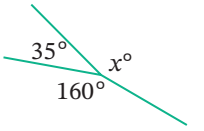
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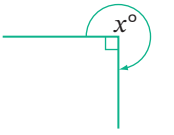
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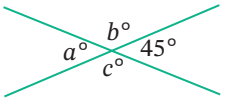
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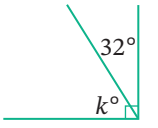
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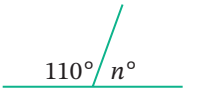
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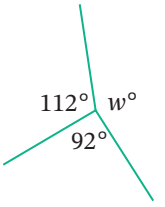
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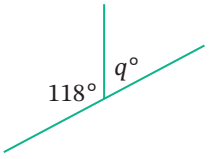
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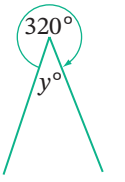
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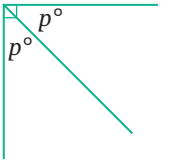
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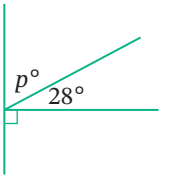
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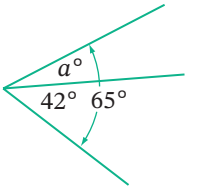
p



q



r

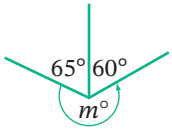


Foundation Standard Complex

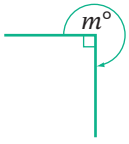
9780170479516

7 Find the value of each variable, giving reasons. R C

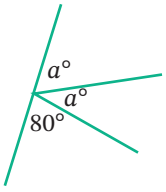
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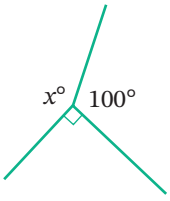
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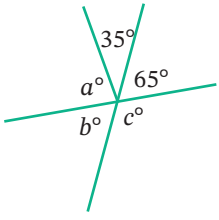
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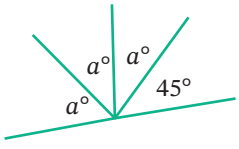
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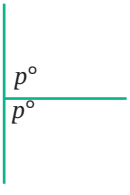
e



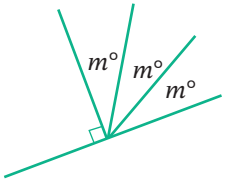
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g

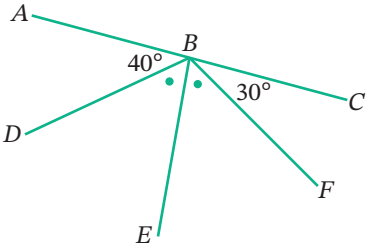


h



8 *ABC* is a straight line and *EB* bisects ∠*DBF*. Find the size of ∠*EBC*.
Select **A**, **B**, **C** or **D**. R

A 95°
B 110°
C 85°
D 140°





Shutterstock.com/Gary Blakeley

Foundation Standard Complex

4.02 Angles on parallel lines

Skillsheet
Angles and parallel lines

Worksheets
Find the unknown angle 1

Investigating angles on parallel lines

Angles in parallel lines

What is the diagram?

Matching angles

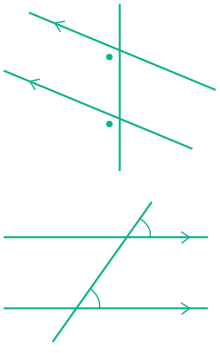
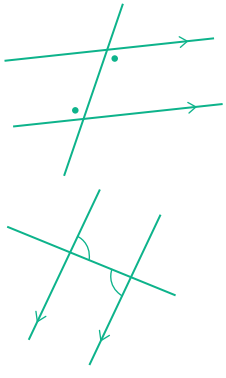
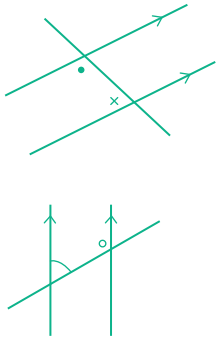
Puzzle
Angles in parallel lines

Videos
Angles on parallel lines

Angle relationships

Video
Angle geometry

When **parallel lines** are crossed by another line (called a **transversal**), special pairs of angles are formed.

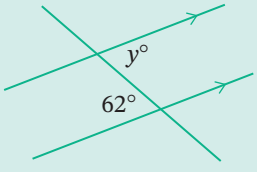
Corresponding angles	Alternate angles	Co-interior angles
Corresponding angles on parallel lines are equal.	Alternate angles on parallel lines are equal.	Co-interior angles on parallel lines are supplementary (add to 180°).
		

- **Corresponding angles** are in 'matching' positions on the *same* side of the transversal: 'corresponding' means 'matching'
- **Alternate angles** are between the parallel lines on *opposite* sides of the transversal: 'alternate' means 'going back and forth in turns'
- **Co-interior angles** are between the parallel lines on the *same* side of the transversal: 'co-interior' means 'together inside'

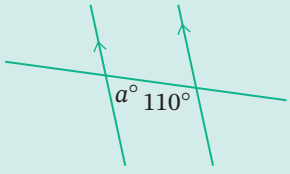
Example 2

Find the value of each variable, giving reasons.

a



b



Solution

a $y = 62$ (Alternate angles on parallel lines)

b $a + 110 = 180$ (Co-interior angles on parallel lines)

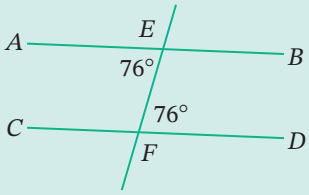
$a = 180 - 110$

$= 70$

9780170479516

Example 3

Prove that the lines AB and CD are parallel.



Solution

$\angle AEF$ and $\angle EFD$ are alternate angles.

$\angle AEF = \angle EFD = 76°$

$\therefore AB \parallel CD$ (Alternate angles are equal)

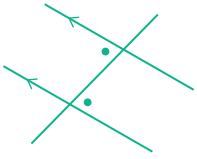
'$\therefore AB \parallel CD$' means 'Therefore line AB is parallel to line CD'

EXERCISE 4.02 ANSWERS ON P. 580

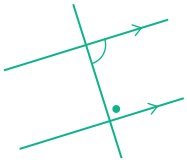
Angles on parallel lines UFRC

1 Is each marked pair of angles corresponding, alternate or co-interior?

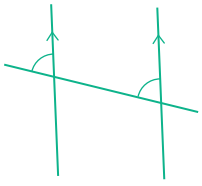
a



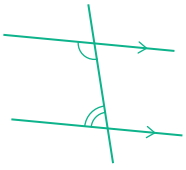
b



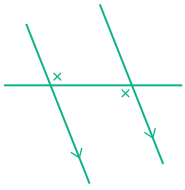
c



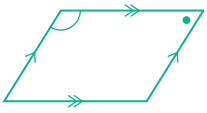
d



e

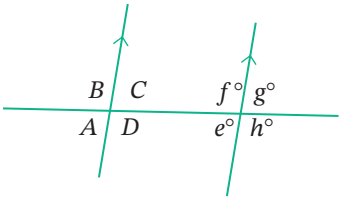


f



2 Which angle is corresponding to the angle marked $g°$?
Select A, B, C or D.

3 For the diagram in question **2**, name an angle that is:



a co-interior to D

b alternate to the angle marked $e°$

c equal to C

d corresponding to B

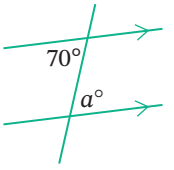
e supplementary to the angle marked $e°$.

Foundation Standard Complex

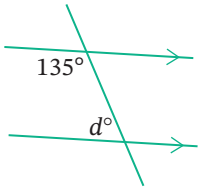
EXAMPLE 2

4 Find the value of the variable in each diagram, giving reasons for your answers. R C

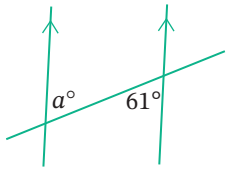
a



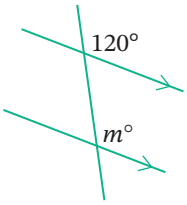
b



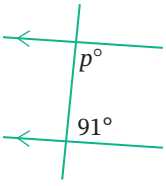
c



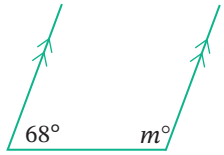
d



e

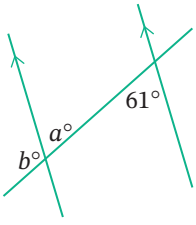


f

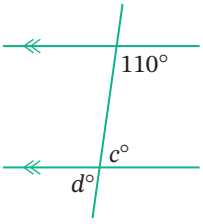


5 Find the value of each variable, giving reasons. R C

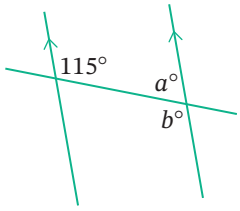
a



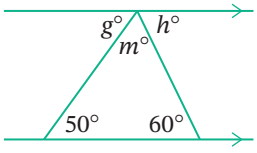
b



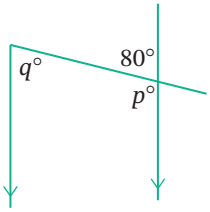
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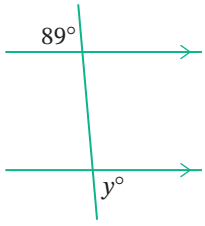
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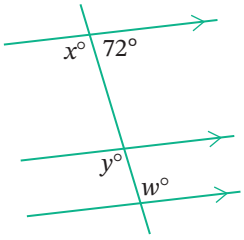
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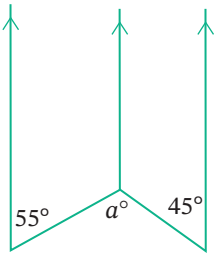
f



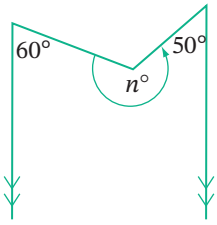
g



h



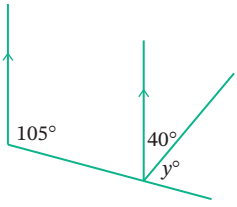
i



6 What is the value of y in this diagram? Select the correct answer **A**, **B**, **C** or **D**. R

A 85 **B** 40

C 35 **D** 65



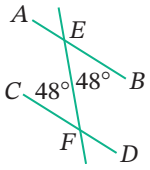
Foundation Standard Complex

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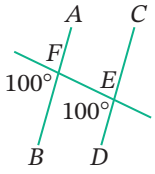
7 For each diagram, decide whether AB is parallel to CD. If it is, then prove it. R C

EXAMPLE 3

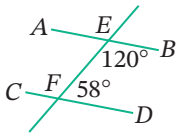
a



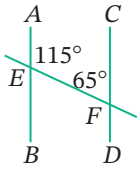
b



c



d

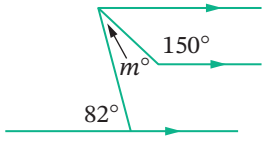


8 Find the value of m in each diagram. R

a



b

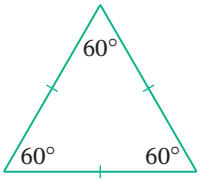
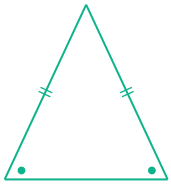
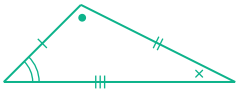


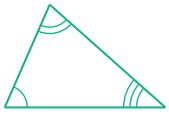
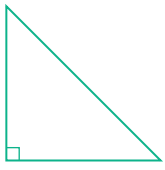
4.03

Classifying triangles

4.03

Triangles can be classified in 2 ways: by their sides or by their angles.

Classifying by sides		
Equilateral triangle	**Isosceles triangle**	**Scalene triangle**
		
3 equal sides (Also 3 equal angles, each 60°)	2 equal sides (Also 2 equal angles, opposite the equal sides)	No equal sides (Also no equal angles)

Classifying by angles		
Acute-angled triangle	**Obtuse-angled triangle**	**Right-angled triangle**
		
3 acute angles (less than 90°)	One obtuse angle (between 90° and 180°)	One right angle (90°)

Skillsheet Naming shapes

Worksheets Properties of triangles

Triangle geometry

Puzzle Classifying triangles

Videos Properties of triangles

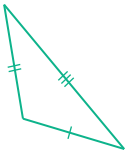
Classifying triangles

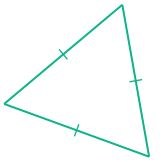
Foundation Standard Complex

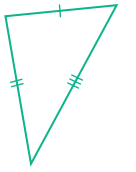
EXERCISE 4.03 ANSWERS ON P. 580

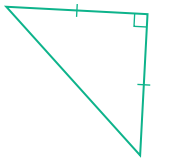
Classifying triangles U F R C

1 Classify each triangle according to its sides and angles. R C

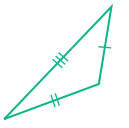
a 

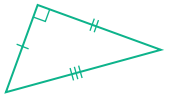
b 

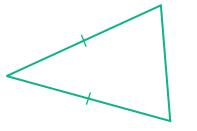
c 

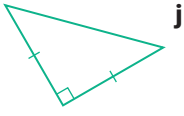
d 

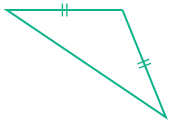
e 

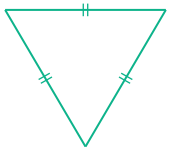
f 

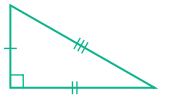
g 

h 

i 

j 

k 

l 

2 Sketch a triangle that is: R C

a right-angled and isosceles

b equilateral

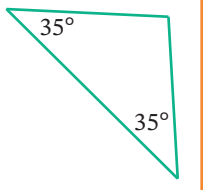
c scalene and obtuse-angled

d acute-angled and scalene

e right-angled and scalene

f acute-angled and isosceles

3 Which type of triangle is this? Select the correct answer **A**, **B**, **C** or **D**. R C



A isosceles and obtuse-angled

B isosceles and acute-angled

C scalene and obtuse-angled

D scalene and acute-angled

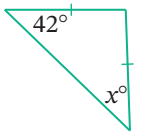
4 Is it possible to draw an obtuse-angled equilateral triangle? Justify your answer. R C

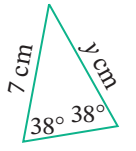
5 Which triangles in question **1** have:

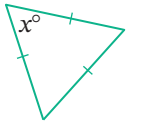
a line symmetry?

b rotational symmetry?

6 Find the value of each variable, giving a reason. R C

a 

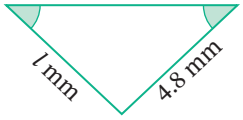
b 

c 

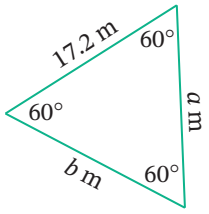
Foundation Standard Complex

9780170479516

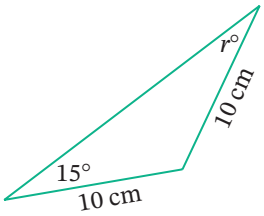
d



e



f



7 Is it possible to draw a triangle with 2 obtuse angles? Why? R C

8 Which triangle is both obtuse-angled and scalene? Select **A**, **B**, **C** or **D**. R C

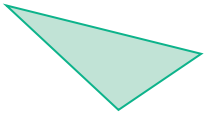
A



B



C



D



9 Copy and complete this table. R C

Triangle	Number of axes of symmetry	Order of rotational symmetry
Equilateral triangle		
Isosceles triangle		No rotational symmetry
Scalene triangle		

Did you know?

It's all Greek or Latin to me!

Many of our words in geometry come from Greek or Latin. Latin was the language of the ancient Roman Empire.

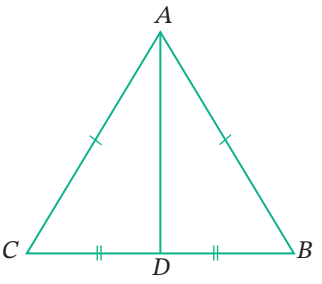
Word	Origin	Meaning
Equilateral	Latin: aequus latus	Equal sides
Equiangular	Latin: aequus angulus	Equal corners
Isosceles	Greek: isos skelos	Equal legs
Scalene	Greek: skalenos	Uneven leg
Acute	Latin: acutus	Sharp
Obtuse	Latin: obtusus	Dull or blunt
Reflex	Latin: reflexus	Bent back
Triangle	Latin: tri angulus	3 corners
Rectangle	Latin: rectus angulus	Right corners
Quadrilateral	Latin: quadri latus	4 sides
Polygon	Greek: poly gonon	Many angles
Diagonal	Greek: dia gonios	From angle to angle
Trapezium/Trapezoid	Latin/Greek: trapeza	Small table

Explain what this sentence means, and illustrate with a diagram:
'A rhombus is equilateral but not equiangular'.

Investigation

The perpendicular bisector in an isosceles triangle

$\triangle ABC$ is an isosceles triangle with $AC = AB$. It has one axis of symmetry, AD.



1 Why is $CD = DB$?

2 Why is $\angle ADC = \angle ADB$?

3 What is the size of $\angle ADC$ and $\angle ADB$?

4 AD bisects side CB. What does 'bisect' mean?

5 $AD \perp CB$. What does '$\perp$' mean?

6 'In an isosceles triangle, the axis of symmetry is the perpendicular bisector of the uneven side.' Explain what this means in your own words.

Quiz
Mental skills 4A

Mental skills 4A: Maths without calculators ANSWERS ON P. 581

Converting fractions and decimals to percentages

To convert a fraction or decimal into a percentage, multiply it by 100%.

1 Study each example.

a $\frac{2}{5} = \frac{2}{5} \times 100\% = \frac{2}{\cancel{5}_{1}} \times \cancel{100}^{20}\% = 2 \times 20\% = 40\%$.

b $\frac{24}{40} = \frac{24}{40} \times 100\% = \frac{\cancel{24}^{3}}{\cancel{40}_{5}} \times 100\% = \frac{3}{\cancel{5}_{1}} \times \cancel{100}^{20}\% = 3 \times 20\% = 60\%$

2 Now convert each fraction to a percentage.

a $\frac{7}{10}$	**b** $\frac{33}{50}$	**c** $\frac{27}{60}$	**d** $\frac{22}{25}$	**e** $\frac{24}{32}$
f $\frac{30}{40}$	**g** $\frac{60}{75}$	**h** $\frac{4}{5}$	**i** $\frac{11}{20}$	**j** $\frac{28}{80}$
k $\frac{15}{50}$	**l** $\frac{16}{20}$	**m** $\frac{54}{60}$	**n** $\frac{18}{40}$	**o** $\frac{13}{25}$

3 Study each example.

a $0.41 = 0.41 \times 100\%$
$= 0.41$
$= 41\%$

b $0.08 = 0.08 \times 100\%$
$= 0.08$
$= 8\%$

c $0.9 = 0.9 \times 100\%$
$= 0.90$
$= 90\%$

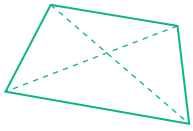
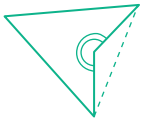
d $0.375 = 0.375 \times 100\%$
$= 0.375$
$= 37.5\%$

4 Now convert each decimal to a percentage.

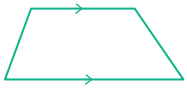
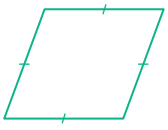
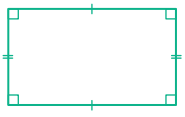
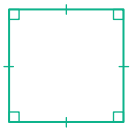
a 0.25	**b** 0.68	**c** 0.17	**d** 0.6	**e** 0.1
f 0.333	**g** 0.59	**h** 0.702	**i** 0.84	**j** 0.7
k 0.428	**l** 0.055	**m** 0.91	**n** 0.7825	**o** 0.314

Classifying quadrilaterals 4.04

A **quadrilateral** is any shape with 4 straight sides. A quadrilateral may be **convex** or non-convex.

Convex quadrilateral	Non-convex quadrilateral
	
• All vertices (corners) point outwards. • All diagonals lie within the shape. • All angles are less than 180°.	• One vertex points inwards. • One diagonal lies outside the shape. • One angle is more than 180° (reflex angle).

There are 6 special types of quadrilaterals.

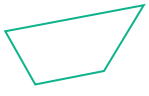
Trapezium	Kite	Parallelogram
 One pair of parallel sides	 2 pairs of equal adjacent sides 'Adjacent' means 'next to each other'.	 2 pairs of parallel sides
Rhombus	**Rectangle**	**Square**
 4 equal sides	 4 right angles	 4 equal sides and 4 right angles

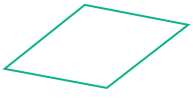
Worksheets
Classifying quadrilaterals
Properties of quadrilaterals
Always, sometimes, never true
Classifying triangles and quadrilaterals
4.04
Puzzles
What shape am I?
Singing in the car
Videos
Properties of quadrilaterals
Classifying quadrilaterals

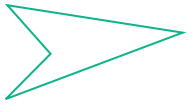
EXERCISE 4.04 ANSWERS ON P. 581

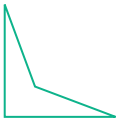
Classifying quadrilaterals U F R C

1 Is each quadrilateral convex or non-convex?

a 

b 

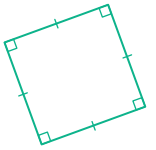
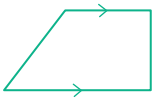
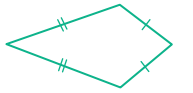
c 

d 

e 

f 

2 Name each quadrilateral. C

a  **b**  **c**  **d** 

3 Name all the quadrilaterals that have: R C

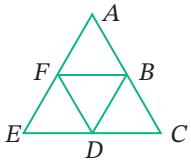
a 4 right angles
b exactly one pair of parallel sides
c 4 equal sides
d opposite sides equal
e opposite sides parallel
f 2 pairs of equal adjacent sides

4 Which quadrilateral is also called a 'diamond'? R C

5 Copy and complete this table.

Quadrilateral	Number of axes of symmetry	Order of rotational symmetry
Rectangle		
Parallelogram		
Trapezium		No rotational symmetry
Rhombus		
Square		
Kite		

6 For this diagram, what type of quadrilateral is: C



a *ACDF*? **b** *FBCD*?

7 A parallelogram is any quadrilateral with both pairs of opposite sides parallel. Which of the following is **not** a special type of parallelogram? Select **A**, **B**, **C** or **D**. R C

A square **B** kite **C** rectangle **D** rhombus

8 Sketch each quadrilateral, showing its main features. C

a rectangle **b** trapezium **c** rhombus **d** kite

9 True or false? (Explain your answers). R C

a A rhombus is a square.
b A square is a rhombus.
c A rectangle is a parallelogram.
d A parallelogram is a quadrilateral with its opposite sides parallel and equal.
e The diagonals of a parallelogram meet at right angles.
f A square is a rectangle.
g A rectangle is a square.

Technology

Properties of quadrilaterals

In this activity, you will use your dynamic geometry software to construct quadrilaterals.

Square

1 Draw a square of length 7.5 cm.

2 Construct the 2 diagonals for the square. Measure the length of each diagonal in the square. What do you notice?

3 Now measure the **size** of each angle of the square. What do you notice?

4 Draw another square with side length 9 cm. Repeat steps **2** and **3**. List 2 properties of the square.

Rectangle

1 Draw a rectangle with sides 6 cm by 3 cm.

2 Measure the length of the diagonals of the rectangle.

3 Now measure the **size** of each angle to check your accuracy. What should the size of each angle be?

4 Draw another rectangle with length 5 cm and width 8.4 cm. Measure the length of each diagonal and the size of each angle.

5 Copy and complete this property: The _____ in a rectangle are _____.

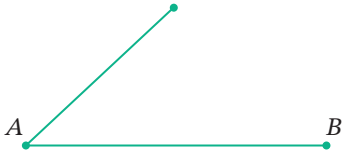
6 In one rectangle, measure the distance from each vertex to the point of intersection of the 2 diagonals. Repeat for the second rectangle. What do you notice?

7 Copy and complete this property: The _____ of a rectangle _____ each other.

Parallelogram

1 Draw an interval 6 cm long. Label it AB.

2 From A, draw the interval AC. Make the interval 4 cm.



3 Complete the parallelogram by drawing an interval from C to D and from B to D.

4 Copy and complete: The _____ sides of a parallelogram are _____.

5 Now measure the size of $\angle CAB$ and $\angle CDB$. Repeat for $\angle ACD$ and $\angle ABD$. What do you notice?

6 Copy and complete: The _____ angles of a parallelogram are _____.

7 Now draw the diagonals of the parallelogram. Measure the length of each diagonal. What do you notice?

8 Copy and complete: The diagonals of a parallelogram are _____.

9 Do the diagonals of a parallelogram bisect each other? Repeat step **6** from 'Rectangle' to help you.

10 Copy and complete: The diagonals of a parallelogram _____ bisect each other.

11 Drag any vertices of the parallelogram that you can. Is it possible to draw other parallelograms with the same dimensions, 6 cm by 4 cm? What do you notice?

12 Use your dynamic geometry software to **accurately** construct other quadrilaterals such as a rhombus, kite or trapezium.



Shutterstock.com/JoeyPhoto

Do you see rhombuses or cubes?

4.05 Properties of quadrilaterals

Worksheets
Properties of quadrilaterals

Diagonal properties of quadrilaterals

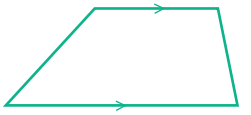
Always, sometimes, never true

Presentation
Angles and shapes

EXERCISE 4.05 ANSWERS ON P. 581

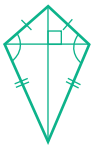
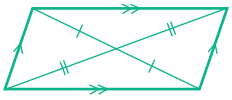
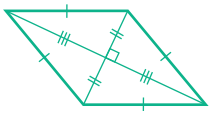
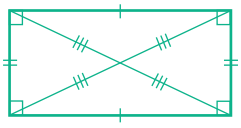
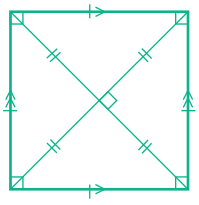
Properties of quadrilaterals U F R C

1 Copy the table below or use the link to print one out. Accurately draw each of the 6 special quadrilaterals below on a sheet of paper, then cut them out. Use your shapes to help you complete the table of quadrilateral properties. R C

a	Trapezium 	• One pair of opposite sides are _____

Foundation Standard Complex

9780170479516

b	Kite 	• 2 pairs of adjacent sides are _____ • One pair of opposite _____ are equal. • Diagonals intersect at _____
c	Parallelogram 	• _____ sides are equal. • Opposite _____ are parallel. • Opposite angles are _____ • Diagonals _____ each other.
d	Rhombus 	• All _____ are equal. • _____ sides are _____ • _____ angles are _____ • Diagonals bisect each other at _____ angles. • Diagonals _____ the angles of the rhombus.
e	Rectangle 	• Opposite sides are _____ • Opposite sides are also _____ • All angles are _____ • Diagonals are _____ • Diagonals _____ each other.
f	Square 	• All sides are _____ • All angles are _____ • Opposite _____ are parallel. • Diagonals are _____ • Diagonals bisect each other at _____

2 Name all special quadrilaterals that have each property. R C

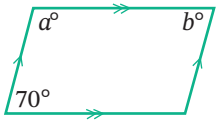
a Opposite sides are equal
b Diagonals cross at right angles
c Opposite angles are equal
d One pair of opposite sides are parallel
e Diagonals bisect each other
f Opposite sides are parallel
g Adjacent sides are of different lengths
h 2 equal diagonals
i All angles are equal
j All sides are equal

3 Copy and complete the working to find the values of a and b. R C

$a + 70 = 180$ (_____ angles on _____ lines)

$a =$ _____

$b =$ _____ (opposite _____ of a parallelogram)



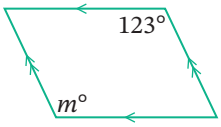
Videos
Properties of quadrilaterals
Classifying quadrilaterals

4.05

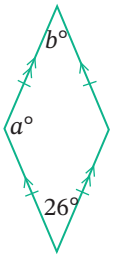
Foundation Standard Complex

4 Find the value of each variable, giving reasons. R C

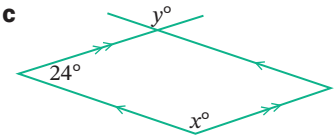
a



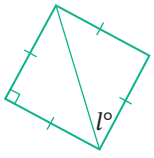
b



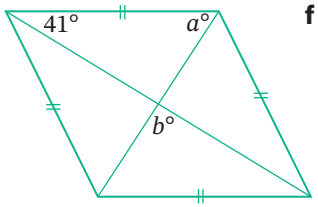
c



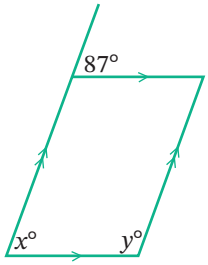
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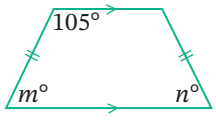
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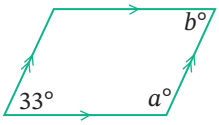
f



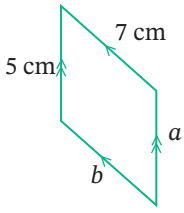
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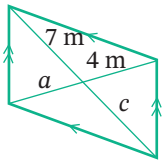
h



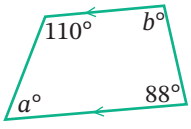
i



j



k



5 I am a quadrilateral with opposite sides equal and parallel.

My diagonals are equal and I have 2 axes of symmetry.

Which quadrilateral am I? Select the correct answer **A**, **B**, **C** or **D**. R C

A parallelogram **B** rectangle **C** square **D** rhombus

6 I am a quadrilateral with opposite sides equal.

My diagonals bisect each other and meet at right angles.

Which quadrilateral am I? Select **A**, **B**, **C** or **D**. R C

A parallelogram **B** trapezium **C** rectangle **D** rhombus

7 A rectangle is a quadrilateral with 4 right angles.

Which one of the following is a special type of rectangle? Select **A**, **B**, **C** or **D**. R C

A square **B** kite **C** parallelogram **D** rhombus

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8 A rhombus is a quadrilateral with all sides equal.

Which one of the following is a special type of rhombus? Select **A**, **B**, **C** or **D**. R C

A square **B** kite **C** parallelogram **D** rectangle

9 The quadrilateral $WXYZ$ is shown. R C



a If $WX = YZ$ and $WZ = XY$, must $WXYZ$ be a rectangle? Why?

b If $\angle WZY = 90°$, must $WXYZ$ be a rectangle? Why?

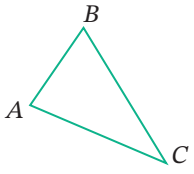
c If the information in parts **a** and **b** are both true, must $WXYZ$ be a rectangle? Why?

Technology

Exterior angle of a triangle

In this activity, you will use dynamic geometry software to discover an important property about the interior and exterior angles of any triangle.

1 Draw a triangle and label each vertex, A, B and C, as shown.

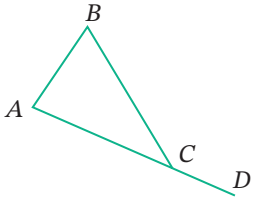


2 Draw a ray from A through C to point D.

3 Find the size of $\angle ACB$, $\angle BAC$ and $\angle BCD$.

4 Calculate $\angle ACB + \angle BAC$.

5 Compare your answer from question **4** with the size of $\angle BCD$. What do you notice?



6 Repeat steps **1** to **5** for 3 other triangles.

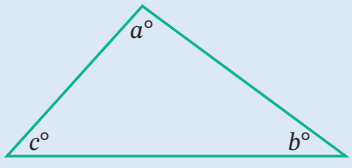
7 Copy and complete: The _____ angle of a triangle is _____ to the sum of the _____ opposite _____ angles.

Angle sums of triangles and quadrilaterals 4.06

Angle sum of a triangle

The angle sum of a triangle is 180°.

$$a + b + c = 180$$



Videos
Angle sums of triangles and quadrilaterals

Angle in polygons

Foundation Standard Complex

Worksheets
Angles in triangles

Find the unknown angle 2

Deductive geometry

Shapes and angles review

Puzzles
Mixed angles

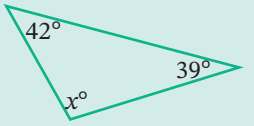
Angles in triangles

Triangles and quadrilaterals

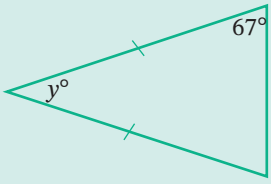
Example 4

Find the value of each variable, giving reasons.

a



b



Solution

a $x + 42 + 39 = 180$ (angle sum of a triangle)

$x = 180 - 42 - 39$

$= 99$

b $y + 67 + 67 = 180$ (angle sum of an isosceles triangle)

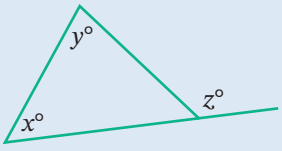
$y = 180 - 134$

$= 46$

Exterior angle of a triangle

The exterior angle of a triangle is equal to the sum of the 2 interior opposite angles.

$$z = x + y$$

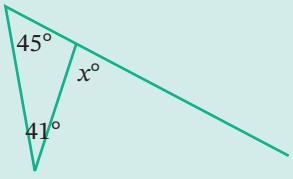


Video
Exterior angle of a triangle

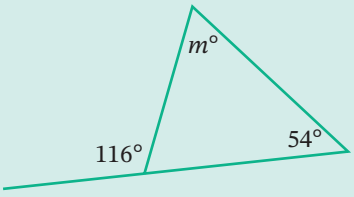
Example 5

Find the value of each variable, giving reasons.

a



b



Solution

a $x = 45 + 41$ (exterior angle of a triangle)

$= 86$

b $m + 54 = 116$ (exterior angle of a triangle)

$m = 116 - 54$

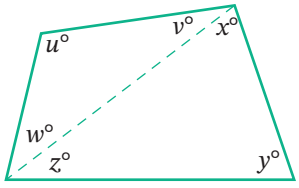
$= 62$

To prove the angle sum of a quadrilateral, we can divide a quadrilateral into 2 triangles along one of its diagonals.
Because the angles in each triangle add to 180°,
the angles in both triangles add to $2 \times 180° = 360°$.

$u° + v° + w° = 180°$ and $x° + y° + z° = 180°$

$\therefore$ Angle sum of quadrilateral $= 180° + 180°$

$= 360°$

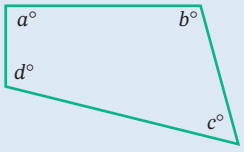


Angle sum of a quadrilateral

The angle sum of a quadrilateral is 360°.

$$a + b + c + d = 360$$

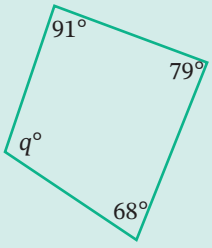
This property is true for both convex and non-convex quadrilaterals.



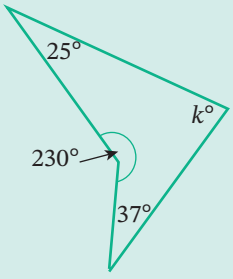
Example 6

Find the value of each variable, giving reasons.

a



b



Solution

a $q + 91 + 79 + 68 = 360$ (angle sum of a quadrilateral)

$q + 238 = 360$

$q = 360 - 238$

$= 122$

b $k + 25 + 37 + 230 = 360$ (angle sum of a quadrilateral)

$k + 292 = 360$

$k = 360 - 292$

$= 68$

Video
Angle sums of triangles and quadrilaterals

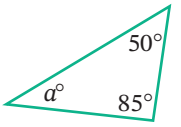
EXERCISE 4.06 ANSWERS ON P. 582

Angle sums of triangles and quadrilaterals UFRC

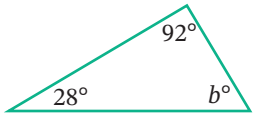
EXAMPLE 4

1 Find the value of each variable. R

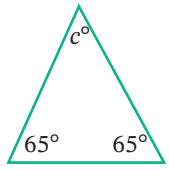
a



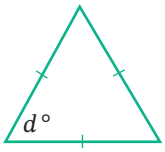
b



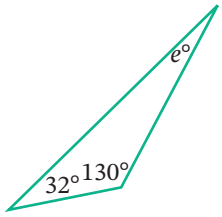
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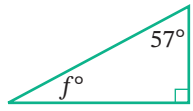
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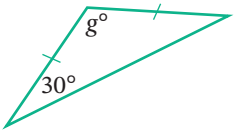
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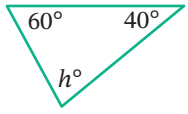
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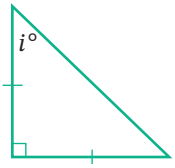
g



h



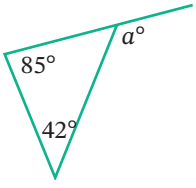
i



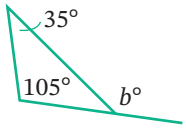
EXAMPLE 5

2 Find the value of each variable. R

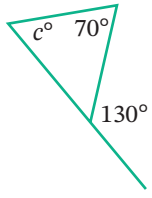
a



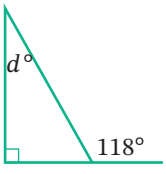
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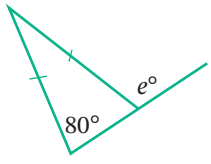
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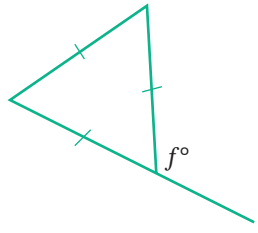
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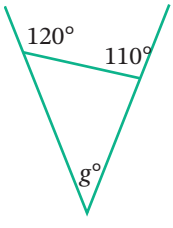
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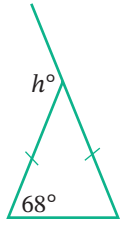
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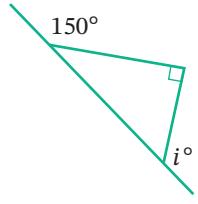
g



h



i



Foundation Standard Complex

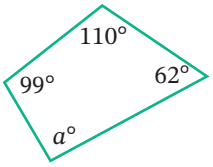
9780170479516

EXAMPLE 6

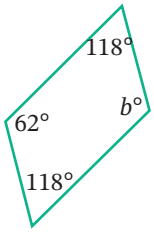
4.06

3 Find the value of each variable. R

a



b



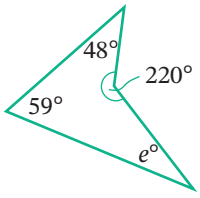
c



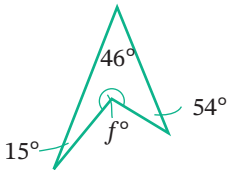
d



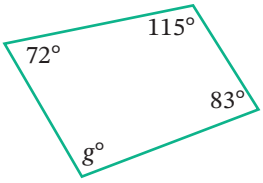
e



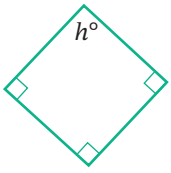
f



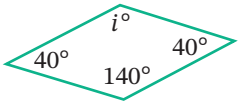
g



h

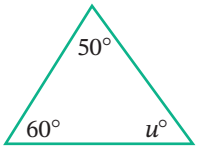


i



4 Find the value of u. Select the correct answer **A**, **B**, **C** or **D**.

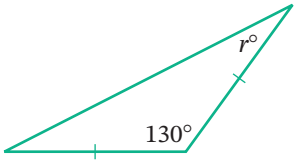
A 50 **B** 60 **C** 70 **D** 80



5 Find the value of r. Select **A**, **B**, **C** or **D**.

A 25 **B** 35

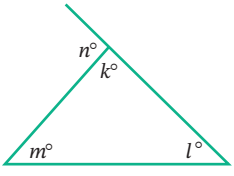
C 50 **D** 100



6 Which equation is correct for this triangle? Select **A**, **B** or **C**. R

A $n = l + m$ **B** $n = k + m$

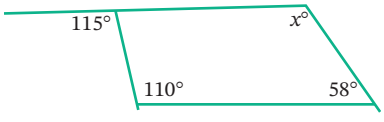
C $n = l + k$



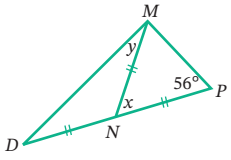
7 Find the value of x. Select **A**, **B**, **C** or **D**.

A 115 **B** 65

C 127 **D** 122



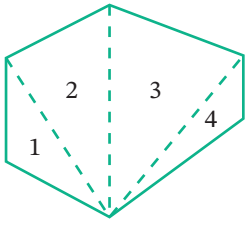
8 Find the size of angles x and y, giving reasons. R C



Foundation Standard Complex

Investigation

Angle sum of a polygon



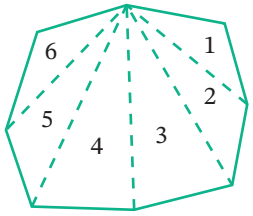
We know that the angle sum of a triangle is 180° and that the angle sum of a quadrilateral is 360°, but how do we find the angle sum of other convex polygons?

A hexagon can be divided into 4 triangles by drawing the diagonals from one vertex.

The sum of the angles in a hexagon = (angle sum of a triangle) × 4

$= 180° \times 4$

$= 720°$



An octagon can be divided into 6 triangles by drawing the diagonals from one vertex.

The total of the angles in an octagon = $180° \times 6 = 1080°$.

1 How is the number of sides related to the number of triangles formed in a shape?

2 Copy and complete these sentences:

The angle sum of a polygon with n sides is

$A = 180 \times$ (number of sides $-$ _____)°

or: $A = 180 \times (n -$ _____)°

3 Find the angle sum of each polygon.

a 11-sided polygon **b** 20-sided polygon **c** 14-sided polygon

4.07 Extension: Angle sum of a polygon

YEAR 9 PATH

A **convex polygon** has all vertices pointing outwards.

A **regular polygon** has all sides the same length and all angles the same size.

Worksheets
Angle sum of a polygon

Angles in regular polygons

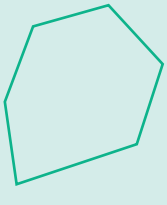
Example 7

These shapes are all hexagons (6-sided).

A



B



C



a Which hexagons are convex?

b Which hexagon is regular?

Solution

a Hexagons **B** and **C** are convex because all of their vertices point outwards.

b Hexagon **C** is regular, because all its sides are equal and all its angles are equal.

Angle sum of a polygon

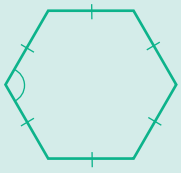
The angle sum of a polygon with n sides is given by the formula:

$$A = 180(n - 2)^\circ$$

This property applies to both convex and non-convex polygons.

Example 8

Find the size of one angle in a regular hexagon.



Solution

A hexagon has 6 sides ($n = 6$).

Angle sum of a hexagon $= 180(6 - 2)^\circ$

$= 180 \times 4^\circ$

$= 720^\circ$

For a regular hexagon:

One angle $= 720^\circ \div 6$

$= 120^\circ$

$\therefore$ Each angle in a regular hexagon is 120°.

EXERCISE 4.07 ANSWERS ON P. 582

Angle sum of a polygon UFRC

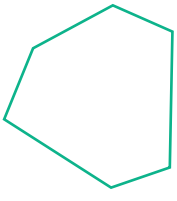
EXAMPLE 7

1 Copy each polygon and write its correct name from this list.

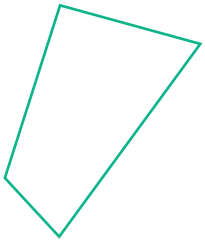
hexagon	nonagon	heptagon	decagon
octagon	triangle	quadrilateral	pentagon

Also, state if each polygon is **regular** or **irregular**, **convex** or **non-convex**. C

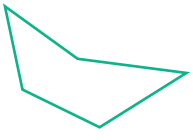
a



b



c



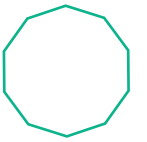
Foundation Standard Complex

YEAR 9 PATH

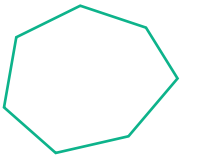
d



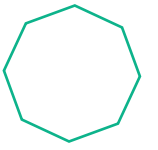
e



f



g



h



2 Copy and complete this table. C

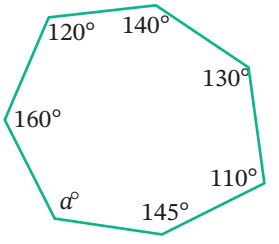
Polygon	Number of sides	Sum of angles inside polygon
hexagon		
heptagon		
octagon		
nonagon		
decagon		

3 Find the sum of the interior angles in:

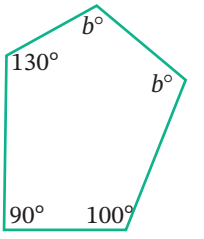
a a 15-agon **b** a 20-agon **c** a 25-agon **d** a 100-agon

4 Find the value of each variable. R

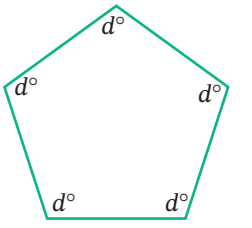
a



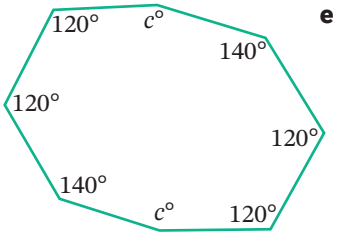
b



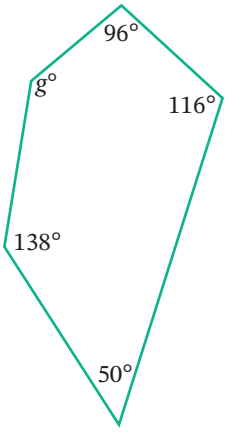
c



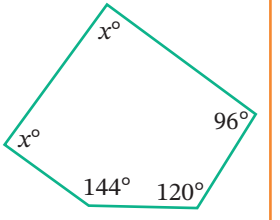
d



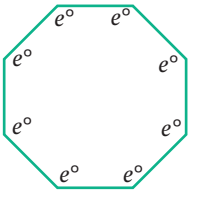
e



f



g



Foundation Standard Complex

9780170479516

5 Find the size of one interior angle in each regular polygon.

a square **b** equilateral triangle **c** regular hexagon
d regular octagon **e** regular decagon **f** regular pentagon
g regular dodecagon

6 Find the number of sides of the polygon that has an angle sum of: R

a 2160° **b** 5760° **c** 4320° **d** 9180° **e** 22 140°

7 Is it possible for a regular polygon to have each interior angle:

a 144°? **b** 130°?

If so, find the number of sides. R

YEAR 9 PATH

EXAMPLE 8

4.08

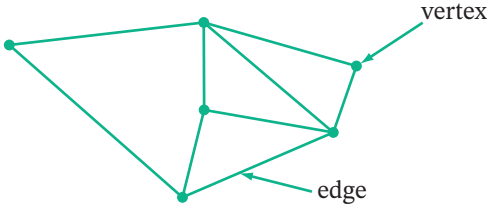
Extension: Networks

4.08

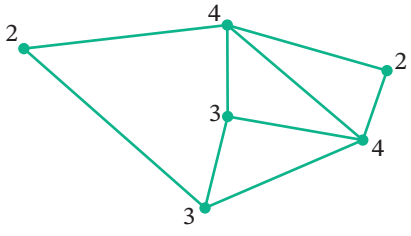
YEAR 9 PATH

A **network** is a collection of objects connected to each other in some way or system. Examples of networks include train networks, electrical circuits, the road traffic network of a city, and social friendship networks such as Instagram.

A **graph** is a diagram of a network where objects are represented by points called **vertices** (or **nodes**), which are joined by lines called **edges**. This graph has 6 vertices and 9 edges.



The **degree** of a vertex is the number of edges connected to it. The degree of each vertex of the graph is shown below.



An **even** vertex has an even number of edges connected to it.

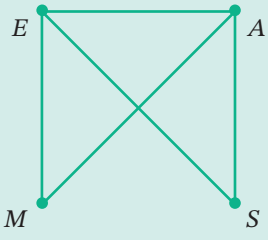
An **odd** vertex has an odd number of edges connected to it.

The graph above has 4 even vertices and 2 odd vertices.

Videos
Networks
The seven bridges of Konigsberg
Networks: Labyrinths and mazes

Example 9

This network shows the friendships between 4 people, Emma (E), Alessio (A), Spyros (S) and Marie (M).



a Find the degree of each vertex in this network.

b How many even vertices are there?

c Show that the sum of the degrees is equal to double the number of edges.

Foundation Standard Complex

YEAR 9 PATH

Solution

a

Vertex	*E*	*A*	*S*	*M*
Degree	3	3	2	2

b There are 2 even vertices (S and M).

c Sum of degrees $= 3 + 3 + 2 + 2 = 10$

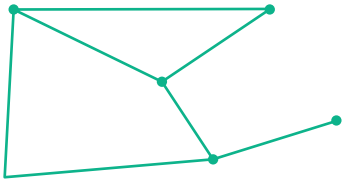
Number of edges $= 5$

Sum of degrees $= 2 \times 5$

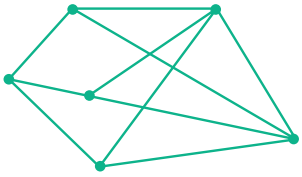
= double number of edges.

Planar graphs

A **planar graph** is a graph hat can be drawn so that no edges cross over. 'Planar' means 2D, on a flat plane, and a planar graph can be drawn on a flat plane so that its edges don't overlap.



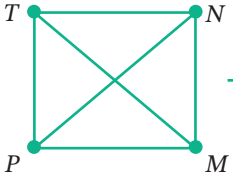
Planar graph



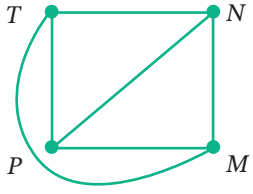
A non-planar graph

Some graphs may have edges crossing but they are still planar graphs because they can be redrawn without any of the edges crossing. For example, Graph **A** below is planar because it can be redrawn as graph **B** or **C**.

A

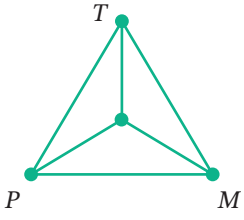


B

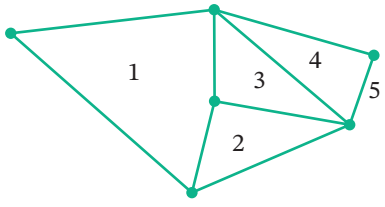


or

C



The **faces** of a network are the regions bounded by edges. When counting the faces of a network, the 'outside' region of the network is also considered a face, called the **infinite face**. This network has 5 faces.



Euler's formula for planar graphs

In 1758, the Swiss mathematician, Leonhard Euler (pronounced 'oiler'), discovered a relationship between the vertices, edges and faces of a planar graph.

Euler is considered the 'father' of network theory.

Euler's formula for planar graphs

For a planar graph,

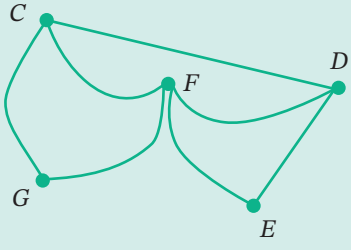
$$v - e + f = 2$$

where v = number of vertices, e = number of edges, f = number of faces

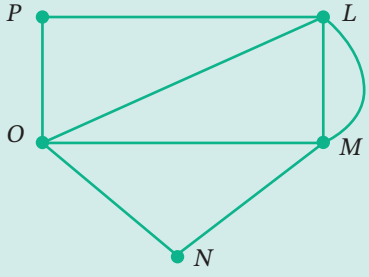
Example 10

Show that each planar graph follows Euler's formula.

a



b



Solution

a $v = 5, e = 7, f = 4$

So $v - e + f = 5 - 7 + 4$

$= 2$

So Euler's formula is true.

b $v = 5, e = 8, f = 5$

So $v - e + f = 5 - 8 + 5$

$= 2$

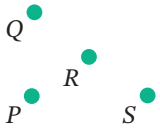
So Euler's formula is true.

EXERCISE 4.08 ANSWERS ON P. 582

Networks U F C

1 Mark, Sue and Alyssa work as a team at a company and email messages to each other each day. Draw a network to represent the movement of the messages. C

2 Towns *P*, *Q*, *R* and *S* are to be joined with roads so that there is a road directly connecting each town to every other town. Copy and complete this road network if the roads must not cross each other. C



EXAMPLE 9

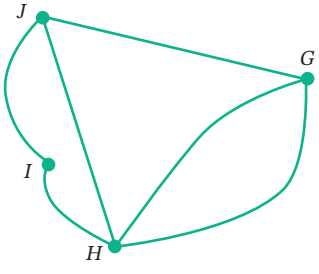
3 For each network:

i find the degree of each vertex

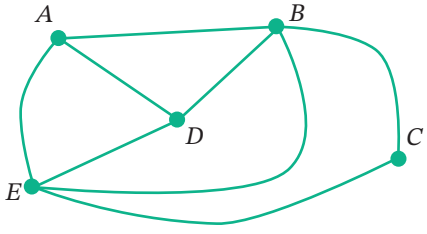
ii state which vertices are even and which are odd

iii show that the sum of the degrees is equal to double the number of edges

a



b



Foundation | Standard | Complex

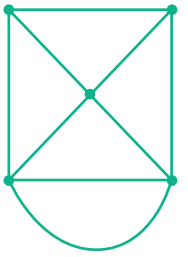
YEAR 9 PATH

EXAMPLE 10

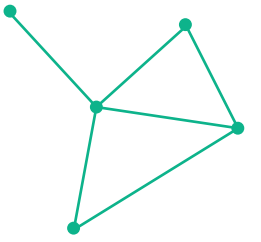
4 Explain why the sum of the degrees of a network is always double the number of edges.

5 For each planar graph, count the number of vertices, edges and faces and check that Euler's formula is true. For graphs **f** and **g**, first redraw one of the edges so that it doesn't cross another edge.

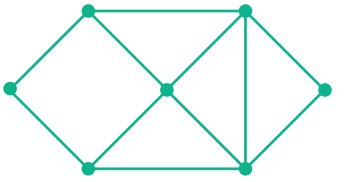
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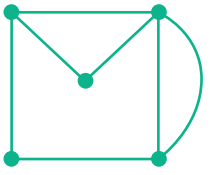
b



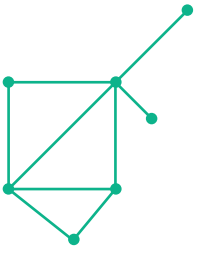
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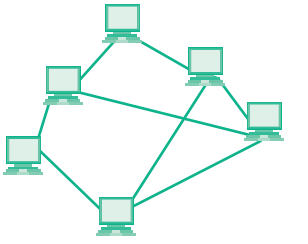
d



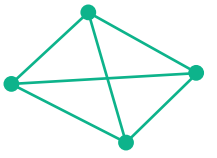
e



f



g



Quiz
Mental skills 4B

Mental skills 4B: Maths without calculators ANSWERS ON P. 582

Converting decimals and percentages to fractions

1 Consider each example.

a $0.35 = \frac{35^{7}}{100_{20}} = \frac{7}{20}$

(2 decimal places, two 0s in the denominator)

b $0.8 = \frac{8^{4}}{10_{5}} = \frac{4}{5}$

(one decimal place, one 0 in the denominator)

c $0.64 = \frac{64^{16}}{100_{25}} = \frac{16}{25}$

d $0.22 = \frac{22^{11}}{100_{50}} = \frac{11}{50}$

2 Now convert each decimal to a fraction.

a	0.75	**b**	0.28	**c**	0.3	**d**	0.14
e	0.06	**f**	0.85	**g**	0.32	**h**	0.49
i	0.56	**j**	0.9	**k**	0.72	**l**	0.65
m	0.2	**n**	0.24	**o**	0.53		

Foundation | Standard | Complex

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3 Consider each example.

a $26\% = \frac{26^{13}}{100_{50}} = \frac{13}{50}$

b $40\% = \frac{40^{2}}{100_{5}} = \frac{2}{5}$

c $8\% = \frac{8^{2}}{100_{25}} = \frac{2}{25}$

d $95\% = \frac{95^{19}}{100_{20}} = \frac{19}{20}$

4 Now convert each percentage to a fraction.

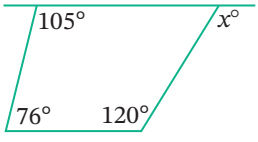
a	76%	**b**	10%	**c**	80%	**d**	45%
e	88%	**f**	56%	**g**	75%	**h**	31%
i	68%	**j**	5%	**k**	60%	**l**	54%
m	6%	**n**	49%	**o**	82%		

Power plus ANSWERS ON P. 582

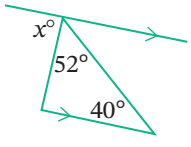
As questions become more complex, it may not be possible to find the answer in one step. It may be necessary to find another angle first.

Find the value of x in each diagram. (Give reasons for all steps.)

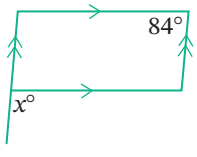
a



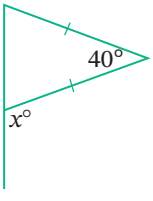
b



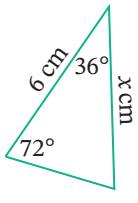
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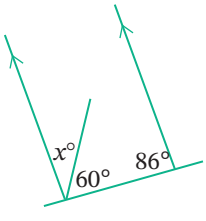
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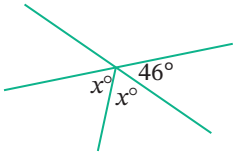
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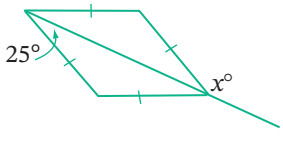
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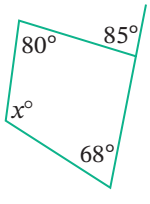
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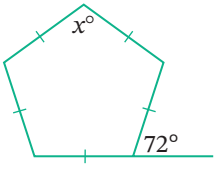
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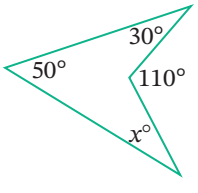
i



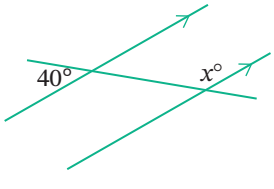
j



k



l



Puzzle Shapes puzzle 1

Worksheet Equal angles

CHAPTER 4 REVIEW

Quiz
Language of maths 4

Puzzle
Geometry crossword

Language of maths

acute	alternate	angle sum	axis/axes
co-interior	complementary	convex	corresponding
diagonal	equilateral	exterior angle	interior angle
isosceles	kite	obtuse	parallelogram
quadrilateral	rectangle	reflex	rhombus
scalene	supplementary	trapezium	vertically opposite

1 What are **supplementary angles**?

2 Name the types of angles associated with parallel lines cut by a transversal.

3 What type of triangle has one angle that is greater than 90°?

4 What is the **angle sum** of a quadrilateral?

5 Illustrate the difference between an **interior angle** and an **exterior angle** of a triangle.

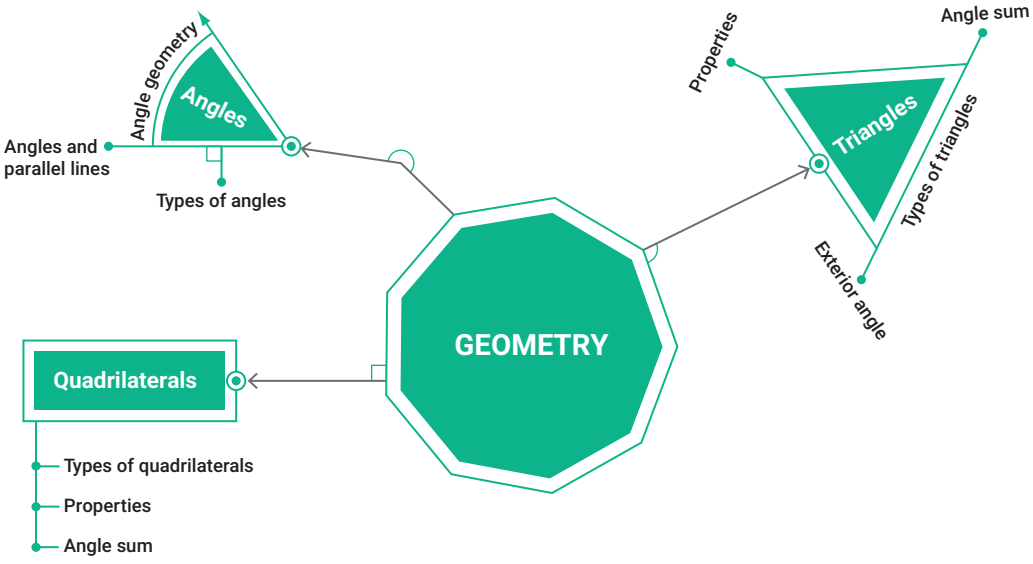
6 What does **equilateral** mean? What is the common name for an 'equilateral parallelogram'?

Topic summary

Worksheet
Mind map: Geometry

- Do you think this chapter is useful? Why? What did you learn in this chapter?
- How confident do you feel with geometry?
- List anything in this chapter that you did not understand. Show your teacher.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 4

ANSWERS ON P. 582

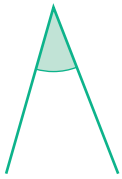
Quiz
Test yourself 4

1 Name each type of angle. 4.01

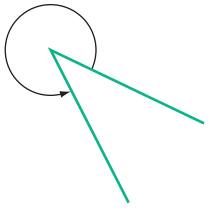
a



b

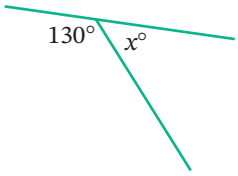


c

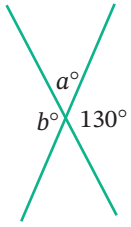


2 Find the value of each variable, giving reasons. 4.01

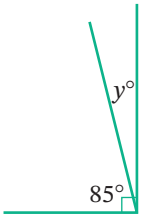
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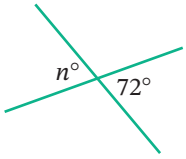
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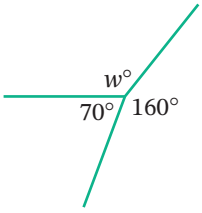
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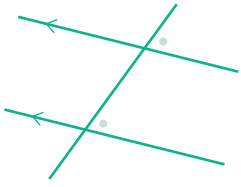


e

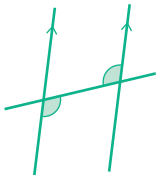


3 Name each type of angle pair. 4.02

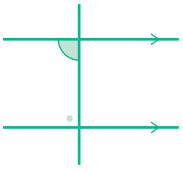
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b

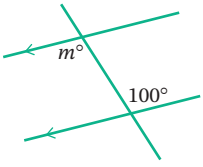


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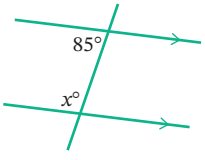


4 Find the value of each variable, giving reasons. 4.02

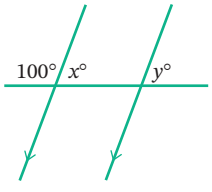
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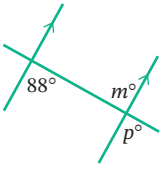
b



c



d



Foundation | Standard | Complex

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4.03

5 Classify each triangle by its sides and angles.

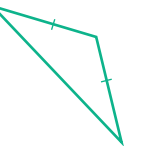
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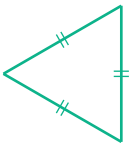
b



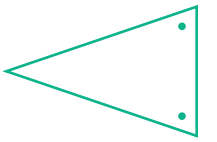
c



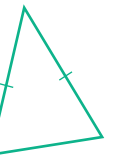
d



e



f



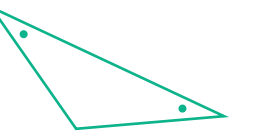
g



h



i



4.04

6 Sketch each quadrilateral.

a parallelogram **b** kite **c** trapezium

4.05

7 Name all the quadrilaterals that have each property.

a All sides equal in length

b No parallel sides

c All angles right angles

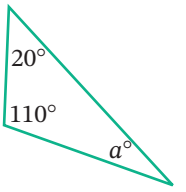
d 2 pairs of opposite angles equal

e Diagonals bisect each other

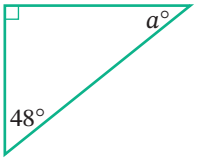
4.07

8 Find the value of each variable.

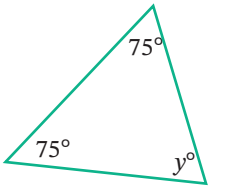
a



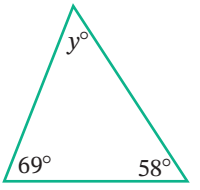
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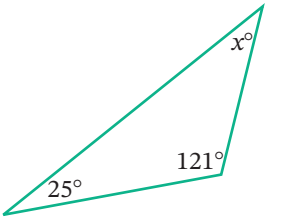
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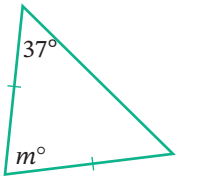
d



e



f

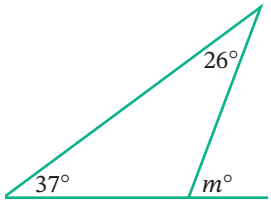


Foundation Standard Complex

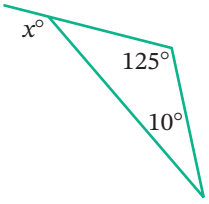
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9 Find the value of each variable. 4.06

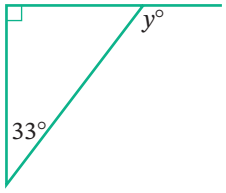
a



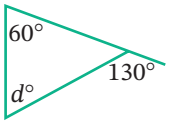
b



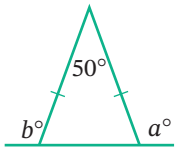
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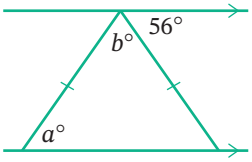
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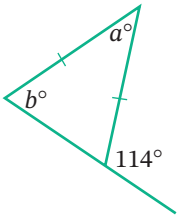
e



f

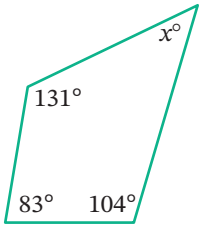


g

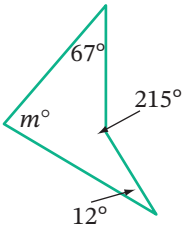


10 Find the value of each variable. 4.06

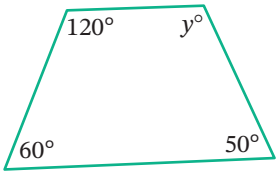
a



b



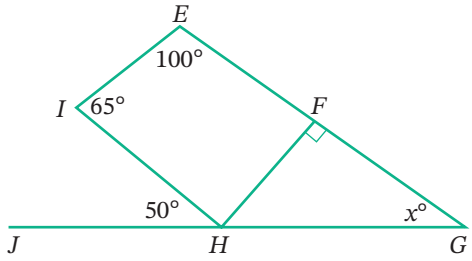
c



11 Find the value of x. Select **A**, **B**, **C** or **D**. 4.06

A 65 **B** 25

C 55 **D** 80



YEAR 9 PATH

12 **a** Find the angle sum of a nonagon. 4.07

b Find the size of one angle in a regular 14-sided polygon, correct to the nearest degree.

c Find the number of sides of the polygon with an angle sum of 3240°.

13 For this network, find: 4.08

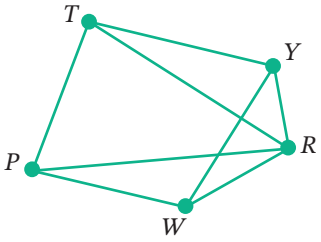
a the number of vertices

b the number of edges

c the degree of T

d the degree of R

e the number of odd vertices



Foundation Standard Complex